

RAG Pipeline

Model Comparison Report

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Pipeline Configuration

Configuration	
Retrieval mode	hybrid
Reranker	OFF
Adaptive weights	ON
Fusion strategy	RRF
BM25 / Sem weights	0.5 / 0.5 (ADAPTIVE)
Filtering	OFF
Cross-modal boost	ON
text_k / rerank_k / image_k	10 / 5 / 5
Embed model	BAAI/bge-large-en-v1.5
Image embed model	ViT-L-14 (laion2b_s32b_b82k)
LLM model(s)	gemma4:e2b
Chunk tokens / overlap	384 / 128

Metric Comparison — All Models

■■ RETRIEVAL (shared across all models) ■■	
M1 Embedding time ↓	12.3186 s
M2 Index size	176 vectors
M3 Retrieval latency ↓	0.0520 s
M4 Fused Relevance (Text)	0.7458
M4 Fused Relevance (Image)	0.4043
M5 Page Coverage@k ↑	95.00 %
M9 Context length	3368 ch
■■ GENERATION QUALITY (per model, vs ground truth) ■■	
M6 ROUGE-1 ↑	0.5138
M7 ROUGE-2 ↑	0.1987
M8 ROUGE-L ↑	0.2915
M10 BLEU ↑	0.1329
M11 METEOR ↑	0.3392
M12 BERTScore ↑	0.8770
M13 FCD ↓	16.96
M14 Faithfulness ↑	92.20 %
M15 GT Coverage ↑	92.02 %
■■ PERFORMANCE (per model) ■■	
M16 E2E Latency ↓	17.3466 s
M17 Throughput ↑	0.058 q/s
M18 CPU Usage ↓	0.28 %
M19 RAM Usage ↓	0.111 GB
GPU avg	26.49 %
GPU peak	100.00 %

* Retrieval metrics are computed once and are identical across model columns because retrieval runs before any LLM is invoked.